

Daniel Kuo

917-530-5715 | dhk3@rice.edu | linkedin.com/in/danielhkuo | github.com/danielhkuo | danielhkuo.github.io

EDUCATION

Rice University

Bachelor of Science in Computer Science, GPA: 3.79

Houston, TX

Aug. 2023 – May 2027

- **Relevant Coursework:** Probability & Statistics, Data Structures & Algorithms, Parallel Computing, Concurrent Programming, Operating Systems, Large-Scale Data Analytics, Data Science, Distributed Program Construction

EXPERIENCE

Goldman Sachs

Software Engineering Intern, Asset & Wealth Management

Dallas, TX

June 2026 – Aug. 2026

- Re-architected the **Java** authentication layer behind a client-onboarding flow serving **35+ client organizations**, moving a statically stored credential out of version-controlled config into a **Sybase**-backed store with scheduled auto-refresh; decoupled rotation from deployment, cutting recovery for a recurring **~8-hour production incident** to a **sub-10-second** DB write.
- Built rotation on **overlapping validity rather than cutover**, refreshing at the **6-month mark** of a **12-month** lifetime with the prior credential kept as a live fallback, so requests authenticate against either credential and rotation has **zero downtime** by construction; made the refresh correct under **concurrent execution across multiple JVMs**.

Labshare

Co-founder & COO (Primary Contributor, Full-Stack Engineering / PM)

Houston, TX

Jan. 2025 – Nov. 2025

- Primary contributor to a lab-inventory platform in **Go** and **React Native**, defining the service layer, **REST** API contracts, and **MongoDB** schema; deployed across **4 AWS regions** behind **CloudFront** and **Nginx** for high availability, cutting inventory overhead **80%** across **30 research labs**.
- Shipped a serverless **AWS Lambda/Step Functions** data-ingestion pipeline in **Go** that parsed heterogeneous lab spreadsheets, used an **LLM** to detect headers and normalize records over goroutine fan-out, and hardened it with chunked calls, exponential-backoff retries, partial-failure tolerance, and deterministic validation of every model output before upsert.

PROS

Software Engineering Intern, Group Sales Optimizer (GSO) Team

Houston, TX

May 2025 – Aug. 2025

- Designed and shipped a historical quote retrieval and replay feature for a B2B price-optimization platform, building a **Java** backend with **Hibernate** and **Oracle** to surface previously internal-only pricing data to clients; collapsed a multi-hour log investigation into a three-click workflow.

PROJECTS

2.5D Parallel Matrix Multiplication | C++, MPI, OpenMP

Apr. 2026

- Implemented a hybrid **MPI + OpenMP** 2.5D matrix multiplication library in **C++**, combining Cannon's algorithm with a hand-tiled, cache-blocked, multithreaded **GEMM** kernel (no BLAS) to reach **503 GFLOPS** across **72 ranks**.
- Profiled a strong-scaling drop to **53%** efficiency with **HPCToolkit** and Jumpshot, traced it to **socket memory-bandwidth saturation**, and rebalanced the MPI rank and OpenMP thread layout to raise efficiency to **72%** and cut runtime by **1.7x**.
- Overlapped communication with computation using persistent non-blocking MPI shifts over a custom Cartesian topology, verified correct to machine epsilon across all valid process-grid configurations.

Synthetic Aperture Tilt-Shift | Python, NumPy, OpenCV, COLMAP, FastAPI, WebGL

Sep. 2026 – Present

- Built a **Python/NumPy** pipeline that renders a handheld phone video into a photograph focused on any tilted plane, recovering camera poses with **COLMAP** and warping frames with **OpenCV** homographies.
- Presented it through a CLI, a **FastAPI** local GUI, and a **WebGL** browser demo, with **29** pytest tests.

Project Plant | Go, Python, Kafka, PostgreSQL, Redis, Docker, k6

Apr. 2026 – June 2026

- Architected a distributed real-time telemetry platform with a **Go** ingestion API, **Kafka** event buffer, concurrent Go stream processor, and **Redis/Postgres** CQRS, sustaining **500+ events/sec** at **<50ms p99 latency** under automated **k6** load tests that gate CI.
- Built a real-time WebSocket dashboard with Redis Pub/Sub fan-out, holding **<100ms p99** reads under write load; drove **TDD** across Go, Python, and TypeScript with testcontainers integration tests.

GPU Hole Compaction | CUDA, C++, NVIDIA CUB

Apr. 2026

- Implemented parallel hole compaction for a plasma-simulation particle array as four data-parallel **CUDA** kernels on an **NVIDIA V100**, compacting **400 million** elements in place **8.2x** faster than the serial CPU baseline with **CUB** prefix sums and no atomics.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Go, Rust, SQL, Haskell

Numerics & Systems: NumPy, OpenCV, MPI, OpenMP, CUDA, multithreading, cache blocking, HPCToolkit profiling, Linux

Data & Infrastructure: PostgreSQL, Oracle, Sybase, MongoDB, Redis, Kafka, AWS, Docker, REST APIs, FastAPI

Testing & Delivery: TDD, pytest, GitHub Actions CI/CD, testcontainers, k6 load testing, code review, Git

AI Tools: Claude Code, Cursor, GitHub Copilot, LangChain, Azure OpenAI, Vertex AI, MCP